CSEC PHYSICS

May/June 2021 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. (a) State Hooke's Law (3 marks)

Hooke's law states that the extension of a spring is directly proportional to the stretching force, provided that the stretching force does not extend the spring beyond its elastic limit. It is given by the formula: $\boldsymbol{F = ke}$

(b) To determine the spring constant (or stiffness) of a spring, a student placed different masses, m, on a hanger that was attached to a spring and recorded the length, I, of the spring. Table 1 shows the mass and corresponding values of length.

TABLE 1: MASS ADDED, m, AND CORRESPONDING Length VALUES, l, OF A SPRING.

Mass, m (kg)	Force, F (N)	$Length, l$ (cm)	Extension, e (cm)
0.00	0.0	70.2	0.0
0.10	1.0	74.4	4.2
0.20	2.0	79.2	9.0
0.30	3.0	83.9	13.7
0.42	4.2	88.2	18
0.50	5.0	92.7	22.5

(i) Complete Table 1 by inserting the missing force and extension values in Columns 2 and 4 respectively. (4 marks)

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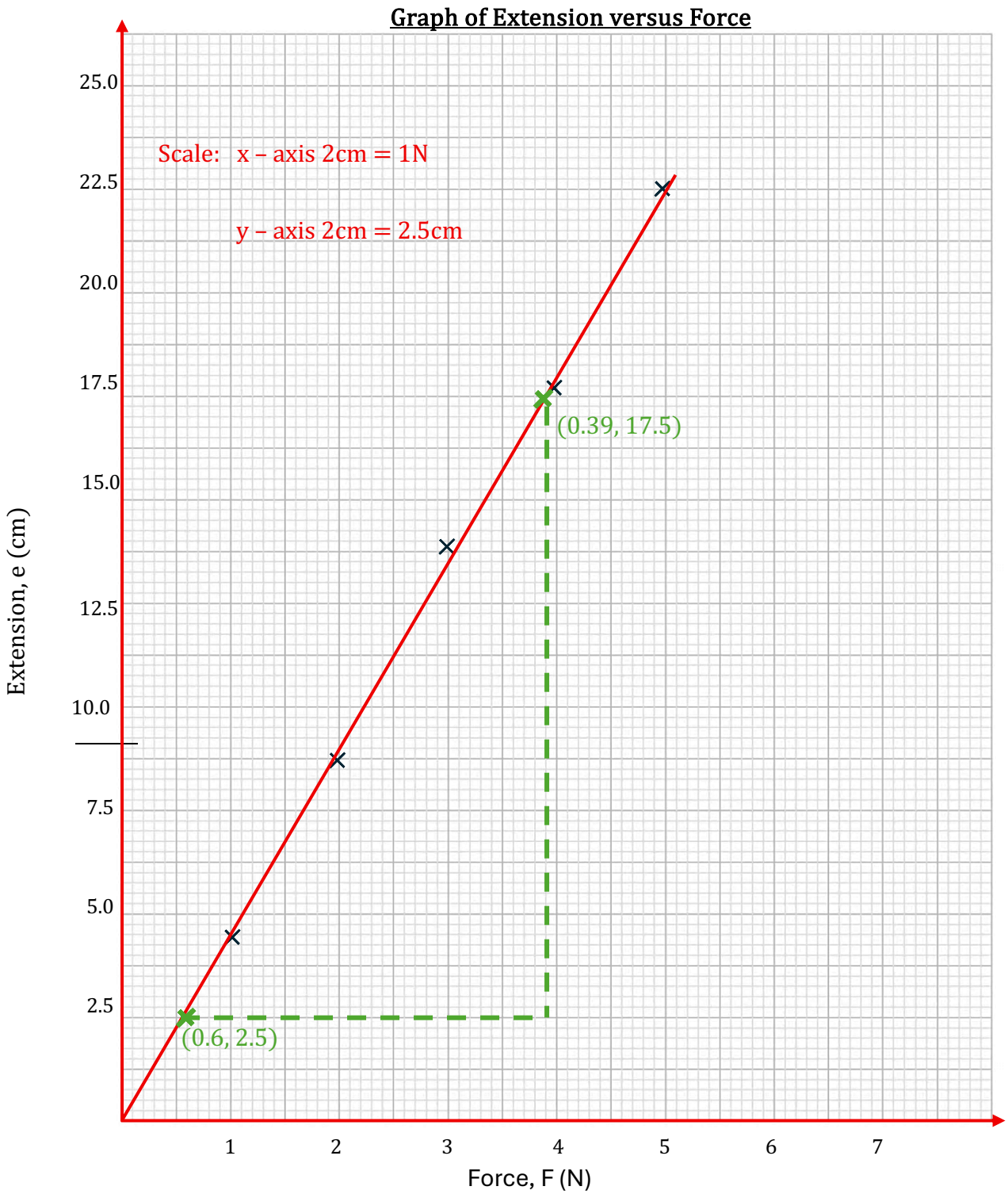
(ii) Using the grid provided in Figure 1 , plot a graph of extension, e vs force, F.

(8 marks)



(c) Calculate the gradient, S, of the graph. (5 marks)

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

Using points (3.9, 17.5) and (0.6, 2.5) from Graph

$$Gradient = \frac{17.5-2.5}{3.9-0.6}$$

$$= 4.55\ cm/N$$

(d) (i) Given that $S = \frac{1}{k}$, calculate the spring constant, k. (4 marks)

$$S = \frac{1}{k}$$

$$k = \frac{1}{S}$$

$$k = \frac{1}{4.55}$$

$$= 0.22N/cm$$

(e) Given that F = ke, calculate the length, l, when a force of 8.0N is applied. Assume that the spring obeys Hooke's Law. (3 marks)

$k = 0.22N/cm \qquad F = 8.0N \qquad e = ?$

$$F = ke$$

$$e = \frac{F}{k}$$

$$= \frac{8}{0.22}$$

$$= 36.4cm$$

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$l = l_o + e$

$= 70.2 + 36.4$

$= 106.6cm$

Total 25 marks

2. (a) Define each of the following terms.

(i) Moment of a force (3 marks)

Moment of a force is the **effect** of a force **about** a fixed point.

It is the product of the magnitude of a force and the perpendicular distance from the pivot to the line of action of the force and is given by the equation: $\boldsymbol{T = Fd}$

(ii) Centre of gravity (2 marks)

The centre of gravity of an object can be defined as the point through which its entire weight acts for any orientation of the object.

(b) State the TWO conditions necessary for a body to be in equilibrium.

- Net forces acting on the body = zero
- Sum of all external moments about the axis must be 0

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(c) Figure 2 shown below is a diagram of a non-uniform wooden plank, AB, which is 2.0m long and weighs 1000N. The plank is supported from a ceiling by two vertical springs, P and Q. The plank is attached at a distance of 0.5 m from each end. Cis the centre of gravity as shown in Figure 2.

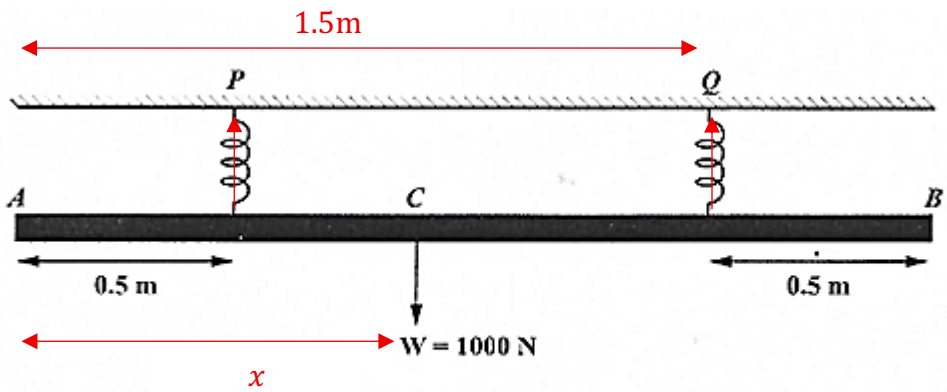

Figure 2. Diagram of a non-uniform wooden plank supported by two springs, *P* and *Q*

When the plank is horizontal, the tension in Spring P is 600N

(i) Calculate the tension in Spring Q. Show all working. (2 marks)

$$Upward\ force = Downward\ force$$

$$P + Q = W$$

$$Q = W - P$$

$$= 1000 - 600$$

$$= 400N$$

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(ii) By taking moments about A, determine the horizontal distance from P to C. (5 Marks)

$$TCWMoments = TACMoments$$

$$1000 \times x = (600 \times 0.5) + (400 \times 1.5)$$

$$1000x = 300 + 600$$

$$x = \frac{900}{1000}$$

$$x = 0.9m$$

Hence, $PC = 0.9 - 0.5$

$$= 0.4m$$

(iii) State why the centre of gravity is located closer to point P than to point Q.

There is more tension in spring P which mean that more of the weight of the plank is concentrated closer to P than Q.

Total 15 marks

3. (a) Waves can be described as transverse or longitudinal.

(i) Explain what is meant by 'transverse wave' (3 marks)

A **transverse** wave is one that has vibrations perpendicular to its direction of propagation. In other words, a transverse wave is where the displacement of particles is at right angles to the direction of travel of the wave.

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(ii) Give one example of a transverse wave (2 marks)

Light Waves

(b) (i) Define the term 'ultrasound' (2 marks)

An ultrasound wave is a sound wave with a frequency that is higher than the maximum frequency that humans can hear. (greater than 20KHz)

(ii) State one example of ultrasound (1 mark)

Medical Imaging

(c) Figure 3 shows the path of a light ray from air through a cube of ice, along with two rays showing partial reflection.

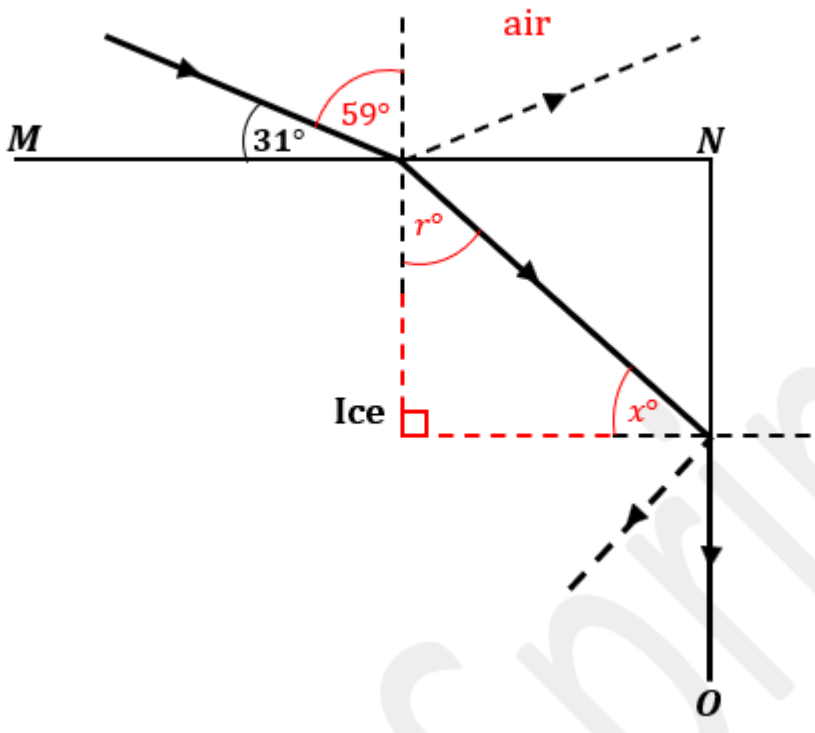

Figure 3. Light ray passing through a cube of ice

Determine

(i) The angle of incidence on the face MN (2 marks)

Angle of incidence $= 90 - 31$

$= 59^0$

(ii) The angle of reflection on the face MN (1 mark)

Law of reflection: i = r

Hence, r = 59^0

(iii) The angle of refraction at the face MN, given that the refractive index of ice is 1.31 (4 marks)

$n_{ice} = 1.31 \qquad r = ?$

Now,

$$n = \frac{\sin i}{\sin r}$$

$$1.31 = \frac{\sin 59}{\sin r}$$

$$\sin r = \frac{\sin 59}{1.31}$$

$$r = \sin^{-1}\left(\frac{\sin 59}{1.31}\right)$$

$r = 40.9°$ (to 1 decimal place)

∴ The angle of refraction on the face MN is 40.9°.

(d) On the diagram in Figure 3, label the critical angle θ, for ice. (1 mark)

See diagram

Total 15 marks

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SECTION B

Answer ALL questions.

4. (a) (i) On Figure 4 below, sketch the graph of pressure versus temperature (in degrees Celsius) for an ideal gas.

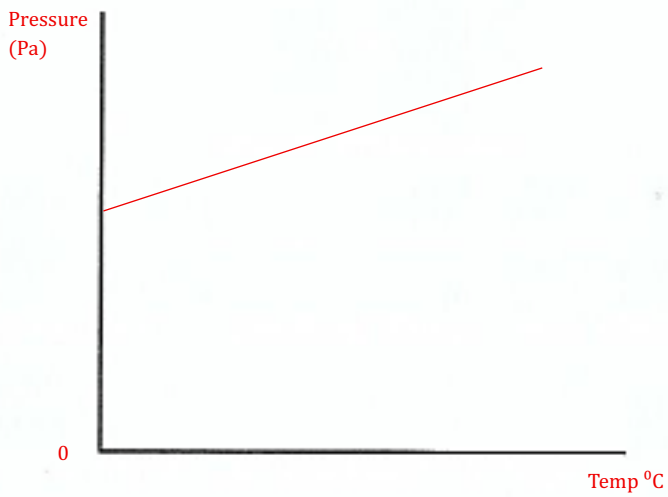

Figure 4. Pressure versus Temperature

(2 marks)

(ii) Explain how the graph could be used to derive the Kelvin scale.

By extending the graph to the left until pressure is equal to zero, the graph will intersect the x axis at Pressure = 0 which is known as absolute zero (-273). This would represent 0K on a kelvin scale and the degree divisions equal the Celsius scale.

(iii) Using T to represent the Kelvin temperature and θ to represent the Celsius temperature, state the mathematical relationship between the Kelvin and Celsius temperature scales. (2 marks)

$$T = \theta + 273$$

(b) A container stores 5.0 m3 of gas at a pressure of 13 atmospheres and a temperature of -23 C. Calculate the volume that the same gas would occupy at a temperature of 27 °C and atmospheric pressure of 1 atmosphere. (5 marks)

$$T_1 = -23 + 273 = 250K$$

$$T_2 = 27 + 273 = 300K$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{(13)(5)}{250} = \frac{(1)V_2}{300}$$

$$V_2 = \frac{(300)(13)(5)}{250}$$

$$= 78m^3$$

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(c) Blocks X and Y in Figure 5 are identical except for their temperatures.

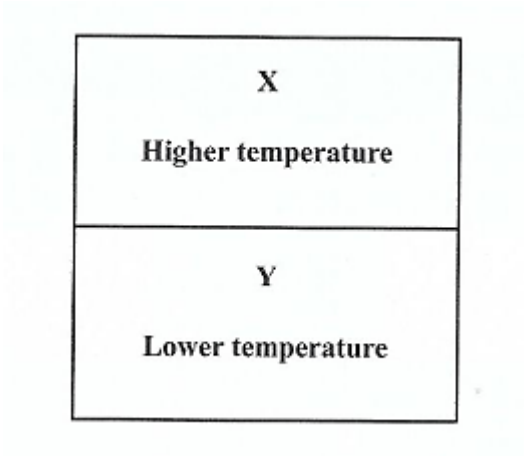

Figure 5.al blocks with different temperatures

(i) Explain the difference in motion of the particles in Block X compared to the motion of particles in Block Y. (2 marks)

X possesses more kinetic energy in its particles as they vibrate more because of the higher temperature

Y possesses less kinetic energy than X so their particles vibrate slower

(ii) In which direction is thermal energy transferred. (1 mark)

$X \rightarrow Y$

(iii) State the condition under which the transfer of thermal energy would cease. (1 marks)

When temperatures of X and Y are the same

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5. Figure 6 shows a simple d.c. motor which consists of a coil of wire, ABCD, between the poles of a magnet. The dotted line represents an axle about which the coil can turn. A current flows through the coil in the direction shown by the arrows.

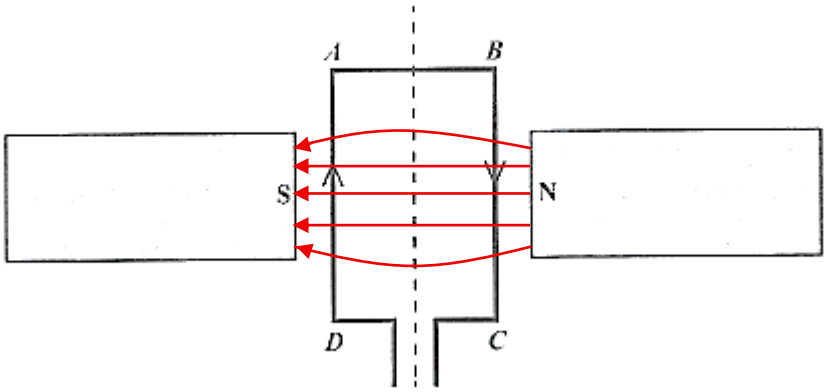

Figure 6. Simple d.c. motor

(a) (i) Draw the magnetic field between the poles of the magnet in Figure 6.

(ii) State the direction in which the side DA of the coil will move as the current starts flowing through the circuit. (1 mark)

Upwards, out of the plane of the paper (Using Fleming's Left Hand rule)

(iii) Name the rule used to determine the direction in which side DA of the coil moves. (1 mark)

Fleming's Left Hand Rule

(iv) Explain why a split ring commutator is used in a dc motor.

The split ring commutator is used in DC motors to reverse the direction of current flow through the motor, every half turn of the coil. This reversal is necessary to ensure the continuous rotation of the motor.

(b) Figure 7 shows a graph of the variation of voltage with time across the output terminals of an a.c. generator.

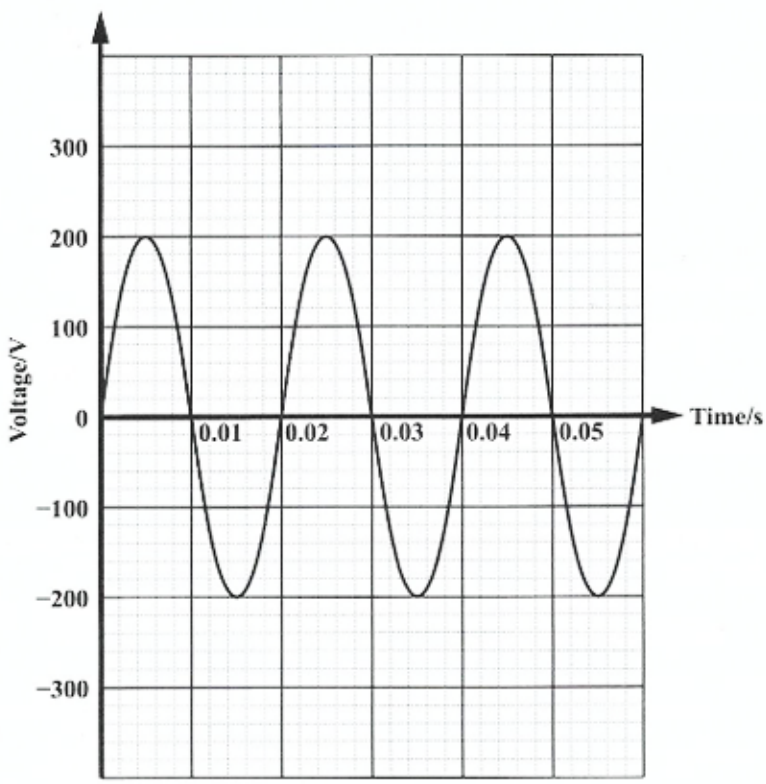

Figure 7. Graph of voltage against time

(i) Calculate the value of the frequency of the generator

$$f = \frac{1}{T}$$

$$T = 0.02\ (from\ graph)$$

$$f = \frac{1}{0.02}$$

$$= 50Hz$$

(ii) The peak voltage is applied by the generator across a 100.0 Ω. Calculate the peak value of the current in the resistor. (3 marks)

$R = 100\Omega \qquad I = ? \qquad V_p = 200V$

$$V = IR$$

$$I = \frac{V}{R}$$

$$= \frac{200}{100}$$

$$= 2A$$

(c) The generator is subsequently rotated at twice the frequency in (b)(i). State the effect this will have on

(i) The period of the output (2 marks)

Since, $f = \frac{1}{T}$

If frequency is doubled, period is halved.

(ii) The value of the output voltage (1 mark)

Voltage will increase

Total 15 marks

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6. Radium -226 is radioactive and is represented by the nuclide $^{226}_{88}Ra$

(a) For a neutral atom of Radium – 226, state

(i) The number of protons

88

(ii) The number of neutrons

138

(iii) The number of electrons

88

(b) A Radium -226 nucleus decays by α-particle emission to Radon (Rn) nucleus.

(i) Explain what is meant by the term 'radioactivity' (3 marks)

Radioactivity is the spontaneous disintegration (decay) of an unstable nucleus

(ii) Complete the equation for the decay of Radium -226.

$$^{226}_{88}Ra \rightarrow \ ^{222}_{86}Rn + \ ^{4}_{2}He^{2+}$$

(c) Radium -226 has a half life of 1600 years.

(i) Define the term 'half-life' (2 marks)

Half life is the time taken for a half of the radioactive nuclei in a sample to decay or to decrease in activity by half.

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(ii) At a certain time, a sample contains 8.0×10^8 Radium nuclei. Calculate the number of α-particles emitted by the Radium nuclei in the next 4800 years.

$$\# \text{ of half lives} = \frac{4800}{1600}$$

$$= 3$$

$$\text{Number of particles} = \left(\frac{1}{2}\right)^{\# \, of \; half \; lives} \times Initial \; number \; of \; nuclei$$

$$= \left(\frac{1}{2}\right)^3 \times 8 \times 10^8$$

$$= 1 \times 10^8 \text{ (left)}$$

$$\text{Therefore } \alpha \; particles = (8 \times 10^8) - (1 \times 10^8)$$

$$= 7 \times 10^8$$ (4 marks)

(iii) How many more time will elapse before the number of Radium nuclei remaining falls to 0.5×10^8? (1 mark)

From 1×10^8 *to* 0.5×10^8 *is one more half life*

Answer: 1600 *years*

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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